CPS 843 (CP 8307) Problem Set 1

Due Monday, September 29th, 11:59pm
(5 assignments in total, final grade is evaluated based on the best 4 out of 5)

Purpose

- Familiar with programming for image processing
- Understand the basic intensity transformation algorithms
- Understand the code and principle for image padding and shearing

Part 1:

Problem 1. Write down the equations for log, inverse log, and power-low transformations and analyze the effect when applying these transformations to an image. Take or download a jpeg image and convert it into grayscale images. Then, perform a power-law transformation on the intensity image with $\gamma = 0.3$ and $\gamma = 3$, respectively. Show the results before and after transformation and make a brief analysis of the results. (4 pts)

Problem 2. Use the same grayscale image generated in Problem 1, show the 8 bit-plane slicing results, then reconstruct an image from the highest 2 and 4 bit-planes, respectively. Show the reconstructed images and make a brief analysis of the results. (4 pts)

Problem 3. For the image and the two transformed images obtained in Problem 1, compute the histogram of the three images, and then apply the histogram equalization to them. For each image, show the histograms and images before and after equalization, then, make a brief analysis of the results. (4 pts)

Problem 4. For the following image demonstrated in class, show the process of histogram equalization, and calculate the probability density function of the equalized image. (4 pts)

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

TABLE 3.1 Intensity distribution and histogram values for a 3-bit, 64 × 64 digital image.

Problem 5. For the following 3-bit image, compute the histogram $p_r(r_k)$ of the image and manually perform histogram equalization. Show the computation steps, as well as the histograms before and after transformation. (4 pts)

1	2	4	7	3
2	4	7	3	1
5	6	2	1	1
4	7	1	1	1

Part 2: (5 pts)

Software package:
The padding and shearing example in Matlab Image Processing Toolbox
https://www.mathworks.com/help/images/padding-and-shearing-an-image-simultaneously.html?prodcode=IP&language=en

Work to do:
- Download or take an image yourself and crop the image properly if necessary.
- Follow the instructions of the four steps and perform the transformation and padding as instructed. You may choose different parameters based on your understanding

Report:
- A brief technical description of each step, the source code, and the results.

Requirements

- The assignment is due on Monday, September 29th @ 11:59pm. You have 24 hours of grace time without penalty. Submissions after the grace period will not be accepted.
- Submit all your work in a single PDF file through D2L (multiple submissions are allowed, but only the last submission will be considered for evaluation).
- Please resize all images properly in line with the text of your report.
- Submit the source code in line with the report of each part.
- You can directly use any available functions of Matlab for your work. Other programming languages and packages are also allowed.
- Complete the report by yourself. We will use Turnitin® for similarity check.
- Suggested report format: IEEE double-column format. The Word and LaTeX template can be found via the link below (please note that any other formats are also allowed)
  http://www.ieee.org/conferences_events/conferences/publishing/templates.html

TA for HW#1:
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You may contact the TA directly if you have any questions about this assignment.